



DDA 3005 — Numerical Methods Algorithms for Nonlinear Equations

Lecture 23

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Conditioning of Nonlinear Equations

Conditioning

The root finding problem is equivalent to evaluating the inverse function

$$f(x^*) = 0 \quad \Longleftrightarrow \quad x^* = f^{-1}(0).$$

Recap: Conditioning

- ▶ The relative condition number of relates the forward and backward error and we have:

$$\text{cond}(f) = \left| \frac{xf'(x)}{f(x)} \right| \implies \text{cond}(f^{-1}) = \left| \frac{f(x)}{xf'(x)} \right|.$$

- ▶ The **absolute condition number** relates the absolute forward and backward error:

$$\begin{aligned} \text{cond}_{\text{abs}}(f) &= |f(x + \Delta x) - f(x)| / |\Delta x| \approx |f'(x)| \\ \implies \text{cond}_{\text{abs}}(f^{-1}) &\approx 1 / |f'(x)|. \end{aligned}$$

- ▶ The root x^* is generally **ill-conditioned** if $f'(x^*) \approx 0$.

Sensitivity & Conditioning



Higher Dimensions and Approximate Solutions

In Higher Dimensions:

- ▶ The condition number of evaluating the function $F(\mathbf{x})$ can be expressed as $\text{cond}(F) = \|\mathbf{x}\| \|DF(\mathbf{x})\| / \|F(\mathbf{x})\|$.
- ▶ The absolute condition number of the root finding problem for the root $\mathbf{x}^*$ of $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is $\|DF(\mathbf{x}^*)^{-1}\|$.
- ▶ The root is **ill-conditioned** if the Jacobian matrix is nearly **singular**.

Approximate Solutions:

- ▶ What do we mean by an “**approximate solution**” $\hat{\mathbf{x}}$ to a nonlinear system:

$$\|F(\hat{\mathbf{x}})\| \approx 0 \quad \text{or} \quad \|\hat{\mathbf{x}} - \mathbf{x}^*\| \approx 0 \quad ?$$

- ▶ The first criterion corresponds to a “small residual”; the second condition measures closeness to a (usually unknown) true solution $\mathbf{x}^*$.

Algorithms for Nonlinear Equations

Methods for Nonlinear Equations

We want to solve nonlinear equations of the form:

$$\text{find } x \in \mathbb{R}: f(x) = 0 \quad \text{or} \quad \text{find } \mathbf{x} \in \mathbb{R}^n: F(\mathbf{x}) = \mathbf{0}.$$

We are going to study / revisit the following methods:

- ▶ Bisection method.
- ▶ General fixed-point methods.
- ▶ Newton's method.

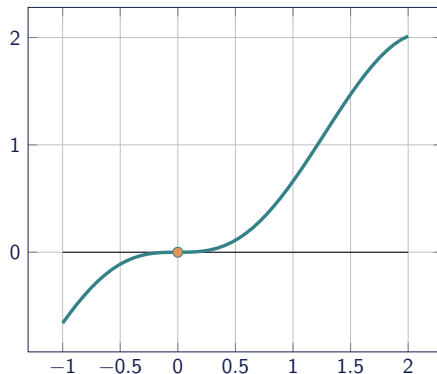
We first consider one-dimensional problems with $f : \mathbb{R} \rightarrow \mathbb{R}$.

Bisection Method in 1D

Bisection Method

Concept and Idea:

- ▶ The **bisection method** starts with an initial interval $[x_\ell, x_r]$ that contains a solution $f(x^*) = 0$.
- ▶ The length of this interval is repeatedly halved until a solution has been isolated with sufficiently high accuracy.



Root Finding Algorithm: Bisection Method

Assume we can find x_ℓ and x_r such that $f(x_\ell) < 0$ and $f(x_r) > 0$.

By the **intermediate value theorem**, there must exist a root of f in $[x_\ell, x_r]$ if f is continuous.

Bisection Method

1. Define $x_m = \frac{x_\ell + x_r}{2}$.
2. If $f(x_m) = 0$, then output x_m .
3. Otherwise:
 - If $f(x_m) > 0$, then let $x_r = x_m$.
 - If $f(x_m) < 0$, then let $x_\ell = x_m$.
4. If $|x_r - x_\ell| < \text{tol}$: stop and output $\frac{x_\ell + x_r}{2}$, otherwise go to step 1.

One can also set the stopping criterion based on $|f(x)| < \text{tol}$.

Bisection Method: Continued

- ▶ The bisection method does not use the function values $f(x)$ to update iterates — only the sign $\text{sign}(f(x))$ is important.
- ▶ Although simple, the bisection method is very useful in practice because it is easy to implement.

Convergence:

- ▶ At each iteration, the length of the interval containing the solution is reduced by $\frac{1}{2}$ — the convergence rate is **linear**.
 - ▶ Given a starting interval $[x_\ell, x_r]$, the length of the interval after k iterations is $(x_r - x_\ell)/2^k$.
- ↪ Achieving the error tolerance `tol` requires

$$\log_2((x_r - x_\ell)/\text{tol})$$

iterations — independent of the function f !

Fixed-Point Iterations in 1D

Fixed-Point Problems

A **fixed-point** of a given function $g : \mathbb{R} \rightarrow \mathbb{R}$ is a scalar x such that

$$x = g(x).$$

- ▶ Many iterative methods for solving nonlinear equations use **fixed-point iteration** schemes of the form:

$$x^{k+1} = g(x^k)$$

where fixed-points of g are solutions to $f(x) = 0$.

- ▶ For a given equation $f(x) = 0$, there may be many equivalent fixed-point problems $x = g(x)$ with different choices for g .

Example: Fixed-Point Problems

Suppose we want to solve $f(x) = x^2 - x - 2 = 0$. Then, fixed points of each of the functions

► $g_1(x) = x^2 - 2$

► $g_2(x) = \sqrt{x+2}$.

► $g_3(x) = 1 + \frac{2}{x}$.

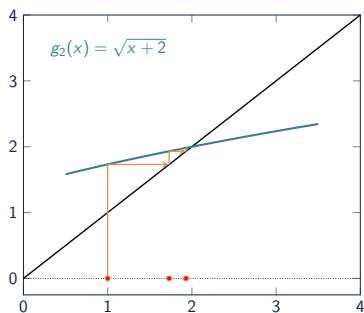
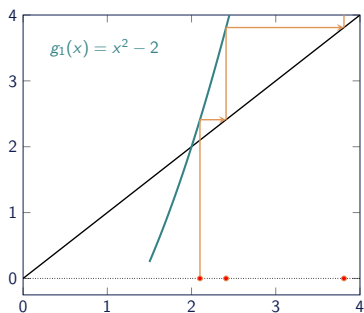
► $g_4(x) = \frac{x^2+2}{2x-1}$.

are solutions to the equation $f(x) = 0$:

$$g_2(x) = x \iff x + 2 = x^2, \quad g_3(x) = x \iff x + 2 = x^2,$$

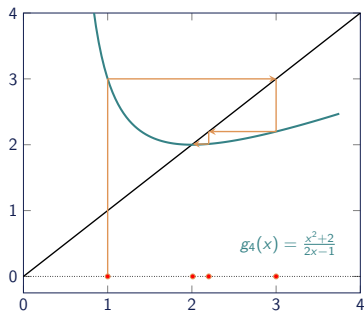
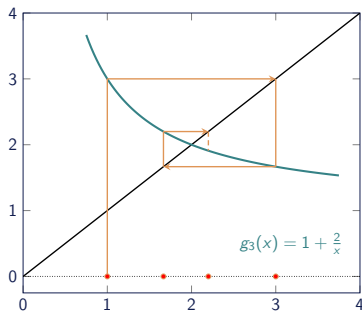
$$g_4(x) = x \iff x^2 + 2 = 2x^2 - x.$$

Visualization: Fixed-Point Iteration — I



- Visualization of the scheme $x^{k+1} = g(x^k)$ for g_1 and g_2 .

Visualization: Fixed-Point Iteration — II



- Visualization of the scheme $x^{k+1} = g(x^k)$ for g_3 and g_4 .

Contractive Mappings

When does this approach converge?

- ▶ Intuitively, applying g to the current iterate x^k should “shrink” or reduce the distance to the fixed-point x^* .
- ▶ This can be guaranteed under certain conditions on g .

Lipschitz Continuity and Contractive Mappings:

A function $g : \mathbb{R} \rightarrow \mathbb{R}$ is **Lipschitz continuous** with constant $L \geq 0$ if

$$|g(x) - g(y)| \leq L \cdot |x - y| \quad \forall x, y \in \mathbb{R}.$$

If $L < 1$, then g is called a **contractive mapping** or **contractive**.

- ▶ If g is contractive, then:

$$|x^{k+1} - x^*| = |g(x^k) - g(x^*)| \leq L \cdot |x^k - x^*| < |x^k - x^*|.$$

Convergence of Fixed-Point Iterations

Convergence of Fixed-Point Methods:

Let $x^* = g(x^*)$ be a **fixed-point** of g and suppose that there is a neighborhood (interval) U containing x^* such that g is **contractive** on U with constant $L < 1$.

- Then, the fixed-point iteration

$$x^{k+1} = g(x^k)$$

converges linearly to x^* with **rate** L for **any initial point** $x^0 \in U$.

Important Question: How to verify whether a mapping is contractive?

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be smooth and let $U \subset \mathbb{R}$ be a given interval. Then, the following conditions are equivalent:

- g is Lipschitz continuous on U .
- We have $|g'(x)| \leq L$ for all $x \in U$.

Remark and Examples

Comments:

- ▶ If x^* is a fixed-point with $|g'(x^*)| < 1$, then we will have $|g'(x)| < 1$ in a neighborhood of x^* (this follows from continuity).
- ▶ If $g'(x^*) = 0$ and g is twice continuously differentiable, then the convergence rate is **at least quadratic**!
- ▶ In the case $|g'(x^*)| > 1$, the fixed-point scheme $x^{k+1} = g(x^k)$ **can not** converge to the fixed-point x^* !

Examples:

- ▶ What can we say about the functions g_1 , g_2 , g_3 , and g_4 ? Are they contractive mappings?

Newton's Method in 1D

Newton's Method

Newton's method is an iterative method. At each point x^k , we first **approximate / linearize** f using a first-order Taylor expansion at x^k :

$$f(x) \approx f(x^k) + f'(x^k)(x - x^k)$$

We set the right-hand side to be 0 and solve it:

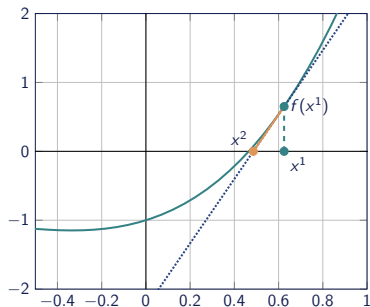
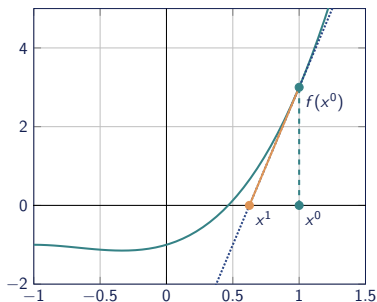
$$x = x^k - \frac{f(x^k)}{f'(x^k)}$$

This x is chosen as our next iterate x^{k+1} .

- Here, we assume $f'(x) \neq 0$ at each step!

Newton's Method: Continued

Newton's method approximates the nonlinear function f near x^k by its **tangent line** $t(x) = f(x^k) + f'(x^k)(x - x^k)$.



Convergence of Newton's Method

Convergence of Newton's Method:

- ▶ Newton's method transforms the nonlinear equation $f(x) = 0$ into a fixed-point problem $x = g(x)$ where

$$g(x) = x - \frac{f(x)}{f'(x)}$$

and hence $g'(x) = f(x)f''(x)/(f'(x))^2$.

- ▶ If x^* is a root with $f(x^*) = 0$ and $f'(x^*) \neq 0$, then $g'(x^*) = 0$.
- ▶ The convergence of Newton's method in such a case is **quadratic**:

$$|x^{k+1} - x^*| \leq C|x^k - x^*|^2$$

for some $C > 0$.

- ▶ **But:** x^0 must be sufficiently close to x^* to ensure convergence!

Algorithms for Nonlinear Systems

Fixed-Point Iterations in $\mathbb{R}^n$

The **fixed-point problem** for $G : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is to find a vector $\mathbf{x} \in \mathbb{R}^n$ such that

$$\mathbf{x} = G(\mathbf{x}).$$

The corresponding **fixed-point iteration** is given by:

$$\mathbf{x}^{k+1} = G(\mathbf{x}^k)$$

Concepts in $\mathbb{R}^n$:

- ▶ Lipschitz continuity can be generalized to n -dimensional mappings.
- ▶ G is **Lipschitz continuous** (on $\mathbb{R}^n$) with constant $L \geq 0$ if

$$\|G(\mathbf{x}) - G(\mathbf{y})\|_2 \leq L \cdot \|\mathbf{x} - \mathbf{y}\|_2 \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n.$$

- ▶ If $L < 1$, then G is said to be **contractive**.
- ▶ The mapping G is Lipschitz continuous on a **convex set** $U \subset \mathbb{R}^n$ if and only if $\|DG(\mathbf{x})\|_2 \leq L$ for all $\mathbf{x} \in U$.

Example and Convergence

The special case $F = \nabla f$:

- ▶ Choosing $F = \nabla f$, we want to find critical points of the optimization problem $\min_{\mathbf{x}} f(\mathbf{x})$.
- ▶ The gradient mapping ∇f is Lipschitz continuous (with constant L) iff the Hessian $\nabla^2 f$ is bounded ($\|\nabla^2 f(\mathbf{x})\| \leq L$ for all $\mathbf{x}$).
- ▶ A typical choice of G : $G(\mathbf{x}) = \mathbf{x} - \alpha \nabla f(\mathbf{x})$.

Convergence:

- ▶ Conditions for convergence are similar to the 1D case.
- ▶ If G is contractive on U , then the fixed-point scheme $\mathbf{x}^{k+1} = G(\mathbf{x}^k)$ converges as long as $\mathbf{x}^0 \in U$.
- ▶ The convergence is typically linear with rate $\eta \approx \|DG(\mathbf{x}^*)\|$.
- ▶ If $DG(\mathbf{x}^*) = \mathbf{0}$, then the convergence is quadratic.

Newton's Method

In n dimensions, **Newton's method** has the form

$$\mathbf{x}^{k+1} = \mathbf{x}^k - DF(\mathbf{x}^k)^{-1}F(\mathbf{x}^k)$$

where $DF(\mathbf{x})$ is the Jacobian matrix of F at $\mathbf{x}$.

- In practice, we do not explicitly invert the matrix $DF(\mathbf{x}^k)$ but instead solve the linear system

$$DF(\mathbf{x}^k)\mathbf{d}^k = -F(\mathbf{x}^k)$$

to obtain the **Newton direction** $\mathbf{d}^k$. The next iterate is then given as

$$\mathbf{x}^{k+1} = \mathbf{x}^k + \mathbf{d}^k.$$

Example: Newton's Method

We consider the nonlinear system

$$F(\mathbf{x}) = 0 \quad \text{with} \quad F_1(\mathbf{x}) = x_1^2 + x_2^2 - 1, \quad F_2(\mathbf{x}) = x_1^4 - x_2^4 + x_1 x_2.$$

The Jacobian of F is given by

$$DF(\mathbf{x}) = \begin{bmatrix} 2x_1 & 2x_2 \\ 4x_1^3 + x_2 & -4x_2^3 + x_1 \end{bmatrix}.$$

For the initial point $\mathbf{x}^0 = [1 \quad 1]^\top$, we then obtain:

$$\mathbf{x}^1 = \mathbf{x}^0 - DF(\mathbf{x}^0)^{-1}F(\mathbf{x}^0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 & 2 \\ 5 & -3 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{16} \begin{bmatrix} 11 \\ 13 \end{bmatrix}.$$

iter	$(\mathbf{x}^k)^\top$	$\ F(\mathbf{x}^k)\ $	$\ \mathbf{x}^k - \mathbf{x}^*\ $
2	[0.548, 0.848]	$4.29 \cdot 10^{-2}$	$2.28 \cdot 10^{-2}$
3	[0.526, 0.851]	$9.37 \cdot 10^{-4}$	$5.18 \cdot 10^{-4}$
4	[0.526, 0.851]	$5.03 \cdot 10^{-7}$	$2.78 \cdot 10^{-7}$
5	[0.526, 0.851]	$1.46 \cdot 10^{-13}$	$8.02 \cdot 10^{-14}$

Convergence of Newton's Method in $\mathbb{R}^n$

Convergence:

- ▶ The convergence rate of Newton's method for nonlinear system is generally **quadratic** provided that the Jacobian matrix $DF(\mathbf{x}^*)$ is nonsingular.
- ▶ The method must be started sufficiently close to a solution to ensure convergence.

Costs of Newton's Method:

- ▶ Computing the Jacobian matrix costs n^2 scalar function evaluations.
- ▶ Solving the linear system requires $\mathcal{O}(n^3)$ computational operations.

Questions?